Secure Student WIL Hours Tracker

Project proposal and problem statement

Cyber Guardians | IncHub Polokwane | CTU WIL programme
Week 2 submission due 16 November 2026
Document Version: 1.0

Executive Summary

The Secure Student WIL Hours Tracker is a cloud-based web application designed to improve the management, verification and reporting of Workplace Integrated Learning hours. The solution aims to replace the already in place manual processes involving paper logbooks and spreadsheets with a secure and centralised platform.

The system will enable students to record WIL activities, upload supporting evidence, monitor progress toward the mandatory 320-hour requirement and submit records for mentor approval. Mentors and programme coordinators will be able to review progress, verify submissions and generate reports through a controlled workflow.

Project overview

We propose a web application that helps students record their Workplace Integrated Learning hours, attach evidence and monitor their progress towards the programme requirement of 320 hours. Assigned mentors will review submitted entries and approve or reject them. Our goal is to make WIL records easier to maintain, verify and prepare for assessment.

This proposal describes the intended system. Features and security controls below are planned and must be implemented and tested before they can be reported as complete.

Problem statement

Students and mentors need a consistent way to record and verify WIL activities. When hours, supporting documents and approvals are kept in separate files or messages, it can be difficult to check progress, identify missing evidence and prepare a complete record for assessment. The project will address this coordination problem by bringing entries, evidence and mentor decisions into one controlled workflow.

During requirements gathering, we will confirm the current process with our mentor and potential users. We will check which difficulties actually occur and adjust the scope to match their feedback. We have not yet measured the frequency of errors or the time spent on the current process.

Target users

- WIL students who need to record activities, submit evidence and view approved hours.
- Assigned workplace mentors who need to review entries and provide feedback.
- Authorised programme coordinators who need an overview of progress and records for assessment.

Project objectives

By the final demonstration, we aim to show a working student-to-mentor submission and approval process, an accurate approved-hours dashboard, protected access to records and an exportable WIL summary using demonstration data.

Proposed solution and scope

A student signs in, records a work session and activity description and attaches supporting evidence where needed.

The entry is submitted to the assigned mentor. The mentor approves it or rejects it with feedback.

Approved entries contribute to the progress total; rejected or pending entries remain visible with their status.

Included in the first version

- Login and role-based access for students, mentors and an authorised coordinator.
- Daily work entries with a date, start and end time, break duration and activity description. Manual entry is the minimum; live clock-in and clock-out is an optional enhancement.
- Evidence upload with defined file types and size limits, linked to the relevant entry.
- Mentor approval or rejection with a reason and a record of the decision.
- Dashboard showing pending and approved hours against the 320-hour target.
- Export of a student's approved hours and activity summary to PDF or CSV.
- Audit records for submissions, edits and approval decisions.

Outside the first version

- Payroll processing
- Human resource management
- Mobile application development
- External company integrations
- Financial management functionality
- Real-time notifications

Rules to confirm with the mentor

- Required evidence formats
- Maximum file upload sizes
- WIL hour validation rules
- Report export requirements
- Attendance verification requirements

Proposed technologies

Software Development

- HTML
- CSS
- JavaScript
- Python (Flask)

Cloud Computing

- Azure App Service
- Azure SQL Database
- Azure Blob Storage

Cybersecurity

- Microsoft Entra ID
- MFA
- RBAC
- Audit Logging
- Azure Key Vault
- POPIA Compliance Controls

Security and privacy requirements

- Restrict each student to their own records. Only assigned, authorised mentors may review their entries; students must never approve their own hours.
- Use an established authentication approach. Protect privileged accounts with MFA where supported; never store plaintext passwords.
- Validate times and calculate hours on the server. Reject negative durations and flag overlapping entries for review.

Proposed success criteria

We will demonstrate and record the following tests. The mentor and team should confirm these criteria before implementation.

Hours calculation. A session from 08:00 to 17:00 with a 60-minute break produces 8 hours.

Approval workflow. A submitted entry can be approved or rejected by its assigned mentor. Rejection requires feedback.

Accurate progress. Only approved entries contribute to the approved-hours total; an approved 8-hour entry adds exactly 8 hours.

Access control. Student A cannot read or change Student B's records or approve any hours through the interface or direct requests.

Evidence protection. An unauthorised request cannot download an attachment; unsupported or oversized uploads are rejected.

Export and traceability. An exported summary matches approved entries, and a submission and approval can be traced to an actor and time.

Key Risks

Unauthorized access to attendance records.

Weak password usage.

Evidence upload misuse.

Data loss.

Failure to meet POPIA requirements.

Project Benefits

Students:

Easier WIL tracking

Improved visibility

Faster report generation

Mentors:

Easier approvals

Better monitoring

Workplace:

Better reporting

Reduced fraud

Improved record management

Team responsibilities

Development. Puleng Machete, Moribishane Ramonyai, Katlego Mojapelo and Kgomoeswana Lethabo will develop the interface, entry calculations, workflow and export features.

Cloud engineering. Kgotso Marapyana will plan hosting, database and storage configuration, deployment and backups.

Cybersecurity. Tshepo Letsoalo and Tshepang Chuene will define security requirements, assess the baseline, maintain risks, review access controls and collect security test evidence.

All members will review the proposal, confirm requirements and participate in integration and demonstrations. The signed team charter remains the record of agreed roles and team working arrangements.

Delivery milestones

- Week 2 — Submit this proposal, confirm the problem and scope with the mentor, complete the security baseline and register the project under Cohort projects. Due 16 November.
- Week 3 — Complete architecture, data flow and security design, including the risk register and threat model. Due 23 November.
- Week 4 — Demonstrate an initial prototype for monitoring review 1. Due 30 November.
- Week 5 — Build and refine the agreed features and assess vulnerabilities. Due 7 December.
- Week 6 — Integrate components and complete functional and security testing. Due 14 December.
- Week 7 — Prepare evidence and demonstrate readiness for monitoring review 2. Due 21 December.
- Week 8 — Deliver the final demonstration, report and reflection. Due 22 December.

Review and submission

Group review date: _____

Mentor feedback or approval reference: _____